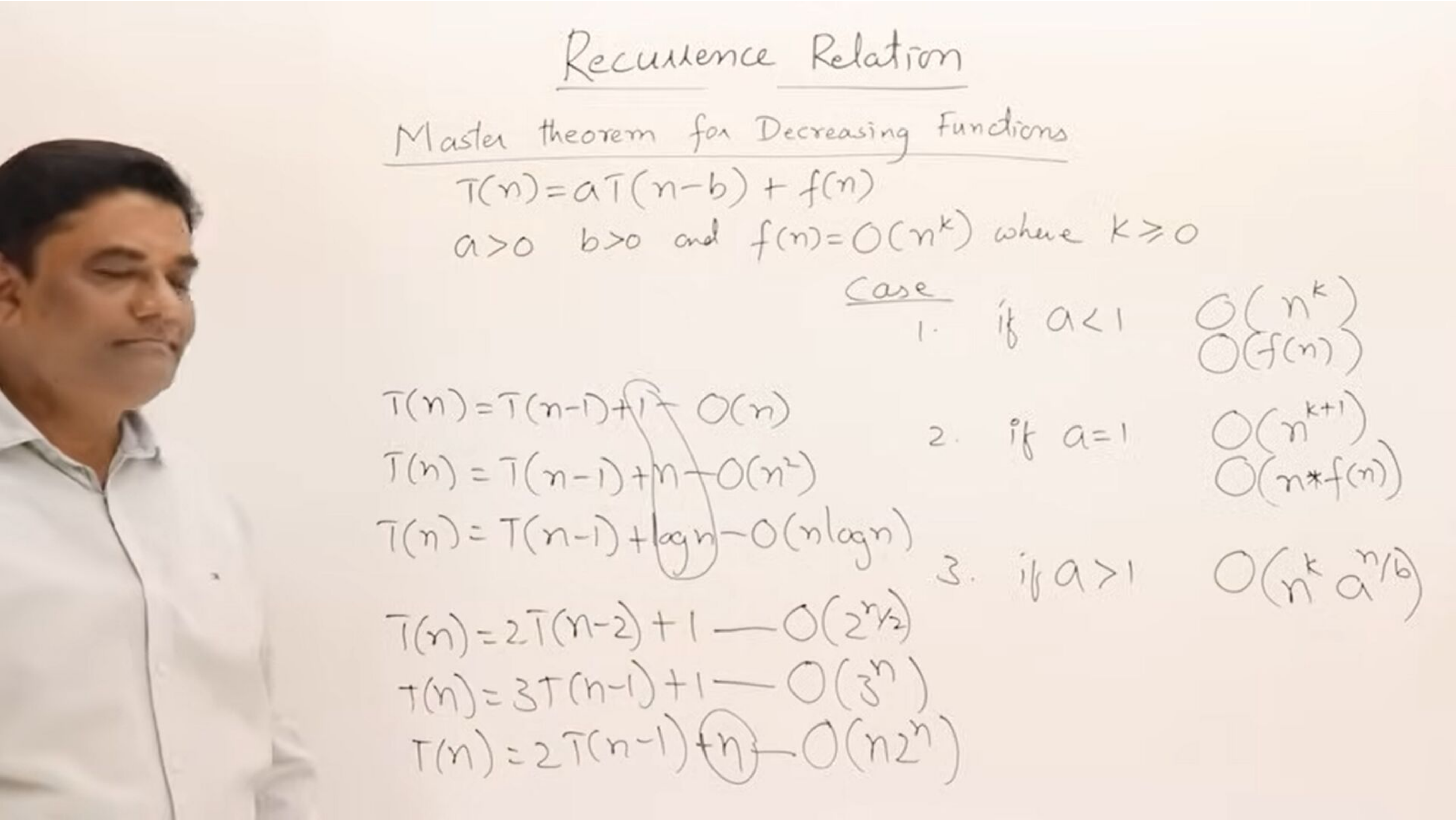
Formula 1


Recurrence Relation
Master theorem for Decreasing Functions
T(n) = aT(n-b) + f(n)
a>0 b>0 and f(n)=O(n^k) where k≥0
Case
1. if a<1 O(n^k)
O(f(n))
2. if a=1 O(n^(k+1))
O(n*f(n))
3. if a>1 O(n^k a^(n/b))
T(n) = T(n-1)+1 — O(n)
T(n) = T(n-1)+n — O(n^2)
T(n) = T(n-1)+logn — O(nlogn)
T(n) = 2T(n-2)+1 — O(2^(n/2))
T(n) = 3T(n-1)+1 — O(3^n)
T(n) = 2T(n-1)+n — O(n2^n)

Formula 2

* Master Method (Theorem)

→ $T(n) = aT(n/b) + f(n)$

$\boxed{a \geq 1}$, $\boxed{b > 1}$

$\cancel{T(n)} = 1.T\left(\frac{n}{1} - 1\right) + 1$

→ Solution is:

$T(n) = n^{\log_b a}\left[U(n)\right]$ ✓

→ $U(n)$ depends on $h(n)$

→ $h(n) = \dfrac{f(n)}{n^{\log_b a}} = \dfrac{n^2}{n^{\log_2 8}} = \dfrac{n^2}{n^3} = \dfrac{1}{n} = n^{-1}$

→ Relation between $h(n)$ and $U(n)$ is:

if $h(n)$	$U(n)$
→ $n^r, r > 0$	$O(n^r)$
→ $n^r, r < 0$	$O(1)$
→ $(\log n)^i, i \geq 0$	$\dfrac{(\log_2 n)^{i+1}}{i+1}$

1) $T(n) = 8T(n/2) + n^2$

2) $T(n) = T(n/2) + C$

$T(n) = 8T(n/2) + n^2$

$a = 8 \quad b = 2 \quad f(n) = n^2$

$T(n) = n^{\log_b a}\, U(n)$

$= n^{\log_2 8}\, U(n)$

$= n^3\, \underline{U(n)}$

$n^3 . 1 = O(n^3)$